

Khovanov homology for singular links and tangles

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November 11, 2023



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- 1 Introduction
 - Knots and links
 - Singular links and their diagrams
- 2 The polynomial invariant
 - Invariants
 - $sl(2)$ singular link extension via webs
- 3 Constructing a stronger invariant
 - Chains and homology
 - Applying a functor and computing homology

Introduction

Acknowledgements

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Knots

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- Because it is difficult to visualize things in $\mathbb{R}^3$, it is convenient to look at projections of these images in $\mathbb{R}^2$.
- We call these projections **knot diagrams**.
- We only consider regular projections of knots, in which only two strands can cross at a time. The double points in the diagram are called **crossings**.

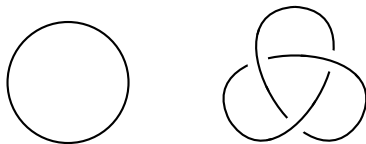


Figure: The unknot and trefoil knot (left to right)

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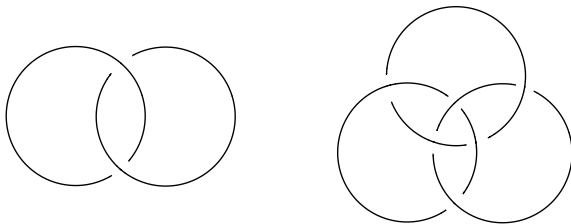


Figure: The Hopf link and the Borromean rings (left to right)

Singular Crossings

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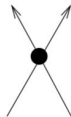


Figure: A singular crossing.

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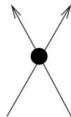


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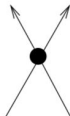


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- In order to define singular links, we first introduce **singular crossings**.
- Recall that we typically don't allow our links to self-intersect (they are embeddings), but singular links are allowed to have finitely many such points of self-intersection, as shown in the figure above.
- Singular link diagrams can have classical crossings just as regular links, and they can also have finitely many singular crossings.

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- Because it is fun!
- As is a standard goal in many areas of topology, we would like to classify these objects.
 - In our case, we wish to classify singular links up to isotopy.
- So what exactly is an isotopy?
- What does it mean for two singular link diagrams to really represent the same singular link? This brings us to the Reidemeister moves (and yes, even more pictures!).

The Reidemeister (type) moves on singular link diagrams

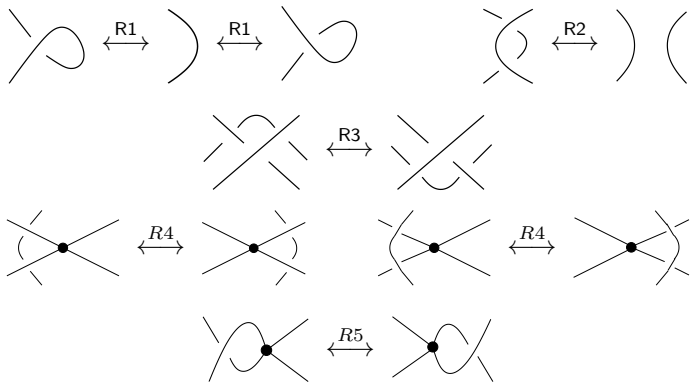


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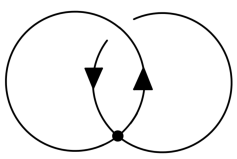


Figure: An oriented singular link diagram

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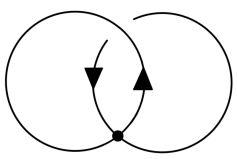


Figure: An oriented singular link diagram

- In this research, we focus on oriented singular links.

The polynomial invariant

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$$f : \{(\text{singular}) \text{ link diagrams}\} \rightarrow \{\text{other mathematical objects}\}.$$

- In the codomain of this map, we often pick objects that are better understood. Objects like groups, rings, modules, etc.
- There are many link invariants that exist, but the codomain for our invariant is the Laurent polynomial ring $\mathbb{Z}[q, q^{-1}]$.

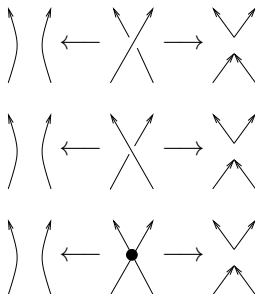
Evaluating States of Singular Link Diagrams

Fix a planar diagram D for the singular link $[D]$.

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Fix a planar diagram D for the singular link $[D]$.

- We replace each crossing in the diagram D with one of the resolutions below.
- After resolving all the crossings of a fixed diagram D , we call the resulting web Γ a *state* of the diagram D .



The evaluation of closed webs

- Because our polynomial invariant will take values in $\mathbb{Z}[q, q^{-1}]$, we need a way to evaluate closed webs.

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$$\langle \bigcirc \cup \Gamma \rangle = (q + q^{-1})\Gamma = \langle \bigcirc \cup \Gamma \rangle$$

$$\langle \text{zigzag} \rangle = \langle \text{arc} \rangle, \quad \langle \text{zigzag} \rangle = \langle \text{arc} \rangle$$

Figure: Web skein relations

Our polynomial invariant for oriented singular links

- Then we can define the polynomial

$$P_2(D) = \sum_{\Gamma} \pm q^{\alpha(\Gamma)} \langle \Gamma \rangle$$

where $\alpha(\Gamma)$ and the sign are determined by the relations below and the sum is taken over all states Γ of D .

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$$\begin{aligned}
 \left\langle \begin{array}{c} \nearrow \searrow \\ \nwarrow \nearrow \end{array} \right\rangle &= q \left\langle \begin{array}{c} \nearrow \\ \nwarrow \end{array} \right\rangle \left\langle \begin{array}{c} \nwarrow \\ \nearrow \end{array} \right\rangle - q^2 \left\langle \begin{array}{c} \nwarrow \nearrow \\ \nearrow \nwarrow \end{array} \right\rangle \\
 \left\langle \begin{array}{c} \nwarrow \nearrow \\ \nearrow \nwarrow \end{array} \right\rangle &= q^{-1} \left\langle \begin{array}{c} \nwarrow \\ \nearrow \end{array} \right\rangle \left\langle \begin{array}{c} \nearrow \\ \nwarrow \end{array} \right\rangle - q^{-2} \left\langle \begin{array}{c} \nwarrow \nearrow \\ \nearrow \nwarrow \end{array} \right\rangle \\
 \left\langle \begin{array}{c} \nwarrow \nearrow \\ \nearrow \nwarrow \end{array} \right\rangle &= (q - q^{-1}) \left\langle \begin{array}{c} \nwarrow \\ \nearrow \end{array} \right\rangle \left\langle \begin{array}{c} \nearrow \\ \nwarrow \end{array} \right\rangle + (q^{-2} - q^2) \left\langle \begin{array}{c} \nwarrow \nearrow \\ \nearrow \nwarrow \end{array} \right\rangle
 \end{aligned}$$

An example

$$P_2 \left\langle \begin{array}{c} \text{Diagram 1} \end{array} \right\rangle = q \left\langle \begin{array}{c} \text{Diagram 2} \end{array} \right\rangle - q^2 \left\langle \begin{array}{c} \text{Diagram 3} \end{array} \right\rangle$$

The diagrams are oriented link diagrams with two components, each represented by a circle with a black dot at the bottom intersection. Arrows indicate the orientation of the links.

- Diagram 1:** Two circles with dots at the bottom intersection. The left circle has an arrow pointing clockwise. The right circle has an arrow pointing counter-clockwise.
- Diagram 2:** Two circles with dots at the bottom intersection. The left circle has an arrow pointing clockwise. The right circle has an arrow pointing clockwise.
- Diagram 3:** Two circles with dots at the bottom intersection. The left circle has an arrow pointing clockwise. The right circle has an arrow pointing counter-clockwise.

$$\begin{aligned}
 &= (q^2 + 1) \left\langle \text{Diagram 1} \right\rangle + (q^{-1} - q^3) \left\langle \text{Diagram 2} \right\rangle \\
 &\quad - (q^3 - q) \left\langle \text{Diagram 3} \right\rangle - (1 - q^4) \left\langle \text{Diagram 4} \right\rangle
 \end{aligned}$$

Diagram 1: A horizontal circle with a vertical loop inside. The vertical loop has two arrows pointing upwards. The horizontal circle has two arrows pointing to the right.

Diagram 2: A horizontal circle with a vertical loop inside. The vertical loop has two arrows pointing downwards. The horizontal circle has two arrows pointing to the right.

Diagram 3: A horizontal circle with a vertical loop inside. The vertical loop has two arrows pointing upwards. The horizontal circle has two arrows pointing to the left.

Diagram 4: A horizontal circle with a vertical loop inside. The vertical loop has two arrows pointing downwards. The horizontal circle has two arrows pointing to the left.

$$\begin{aligned}
 &= (q^2 + 1) \left\langle \text{Diagram 1} \right\rangle + (q^{-1} - q^3) \left\langle \text{Diagram 2} \right\rangle \\
 &\quad - (q^3 - q) \left\langle \text{Diagram 3} \right\rangle - (1 - q^4) \left\langle \text{Diagram 4} \right\rangle \\
 &= (q^2 + 1)(q + q^{-1}) \left\langle \text{Diagram 5} \right\rangle + (q^{-1} - q^3) \left\langle \text{Diagram 6} \right\rangle - (q^3 - q) \left\langle \text{Diagram 7} \right\rangle \\
 &\quad - (1 - q^4) \left\langle \text{Diagram 8} \right\rangle
 \end{aligned}$$

The diagrams are as follows:

- Diagram 1: A single circle with two internal loops, one clockwise and one counter-clockwise.
- Diagram 2: A single circle with two internal loops, one clockwise and one counter-clockwise, with an additional crossing.
- Diagram 3: A single circle with two internal loops, one clockwise and one counter-clockwise, with an additional crossing.
- Diagram 4: A single circle with two internal loops, one clockwise and one counter-clockwise, with an additional crossing.
- Diagram 5: Two separate circles, one clockwise and one counter-clockwise.
- Diagram 6: Two separate circles, one clockwise and one counter-clockwise, with an additional crossing.
- Diagram 7: Two separate circles, one clockwise and one counter-clockwise, with an additional crossing.
- Diagram 8: Two separate circles, one clockwise and one counter-clockwise, with an additional crossing.

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 &= (q^2 + 1)(q + q^{-1})^2 + (q^{-1} - q^3)(q + q^{-1}) - (q^3 - q)(q + q^{-1}) - (1 - q^4)
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$$\begin{aligned}
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&\quad - (1 - q^4) \left\langle \text{Diagram 8} \right\rangle \\
&= (q^2 + 1)(q + q^{-1})^2 + (q^{-1} - q^3)(q + q^{-1}) - (q^3 - q)(q + q^{-1}) - (1 - q^4) \\
&= 2q^2 + 2q^{-2} + 5.
\end{aligned}$$

Constructing a stronger invariant

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- We aim to construct an analogous categorification for the P_2 polynomial which was just introduced.
- The resulting homology theory for singular links when restricted to classical links should be isomorphic to the original Khovanov homology.

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 - the homology groups of the complex are invariants of the singular link and
 - the graded Euler characteristic of the chain complex is the polynomial P_2 .

Chain Complexes and Homology

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Definition

A **chain complex** (X^r, d^r) is a sequence of group homomorphisms between abelian groups

$$\dots \longrightarrow X^{r-1} \xrightarrow{d^{r-1}} X^r \xrightarrow{d^r} X^{r+1} \xrightarrow{d^{r+1}} \dots$$

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Definition

Given a chain complex (X^r, d^r) of abelian groups and group homomorphisms, we define the **r -th homology group** of the complex as the quotient

$$H^r(X) := \ker(d^r) / \operatorname{im}(d^{r-1}).$$

Constructing a chain complex of webs and foams

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$$\text{Crossing} = \left[0 \rightarrow \begin{array}{c} \text{X} \\ -1 \end{array} \xrightarrow{\{2\}} \begin{array}{c} \text{Diagram with red line} \\ 0 \end{array} \left(\{1\} \rightarrow 0 \right) \right]$$

The diagram shows a sequence of moves in the Reidemeister theorem proof. It starts with a crossing of two strands, followed by an equals sign and a bracketed sequence of moves. The first move is a strand with a loop labeled '0'. The second move is a strand with a loop labeled '-1'. The third move is a crossing of two strands labeled '1'. The final move is a strand with a loop labeled '-2' followed by '0'. A red wavy line connects the second and third moves, and a dashed line connects the third and fourth moves.

$$\bullet = \left[0 \longrightarrow \left(\begin{array}{c} \text{hook} \\ \text{hook} \end{array} \right) \left(\begin{array}{c} \text{hook} \\ \text{hook} \end{array} \right) \xrightarrow{D} \left(\begin{array}{c} \text{hook} \\ \text{hook} \end{array} \right) \left(\begin{array}{c} \text{hook} \\ \text{hook} \end{array} \right) \longrightarrow 0 \right]$$

This chain complex is actually an invariant (in progress)

Conjecture

The chain complex is an oriented singular link invariant which recovers the P_2 polynomial after a certain structure-preserving functor is applied. The P_2 polynomial is equal to the graded Euler characteristic of the resulting algebraic chain complex.

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- So where does the homology come in, and why is homotopy equivalence important?
 - Well, homology is a homotopy invariant.

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- The resulting homology groups should be invariants for the singular links. The graded Euler characteristic of the chain complex will recover the $P_2(L)$ polynomial that was described above.

$$P_2(D) = \sum_{i,j \in \mathbb{Z}} (-1)^i q^j \text{rank}(H^{i,j}(D))$$

Thank You!

Questions?

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Bonus: Chain Homotopy

Definition

Let $\mathcal{C}$ be a category. We say that two morphisms $F, G : (X^r, d_X^r) \rightarrow (Y^r, d_Y^r)$ in $\text{Kom}(\mathcal{C})$ are **homotopic** if there exist “backwards diagonal” morphisms $h^r : X^r \rightarrow Y^{r-1}$ such that $F^r - G^r = h^{r+1}d^r + d^{r-1}h^r$ for all r . We denote this as $F \simeq G$.

$$\begin{array}{ccccccc}
 \dots & \longrightarrow & X^{r-1} & \xrightarrow{d_X^{r-1}} & X^r & \xrightarrow{d_X^r} & X^{r+1} \longrightarrow \dots \\
 & & \downarrow F^{r-1}, G^{r-1} & \swarrow h^r & \downarrow F^r, G^r & \swarrow h^{r+1} & \downarrow F^{r+1}, G^{r+1} \\
 \dots & \longrightarrow & Y^{r-1} & \xrightarrow{d_Y^{r-1}} & Y^r & \xrightarrow{d_Y^r} & Y^{r+1} \longrightarrow \dots
 \end{array}$$

Definition

Let (X^r, d_X^r) and (Y^r, d_Y^r) be chain complexes and $F : (X^r, d_X^r) \rightarrow (Y^r, d_Y^r)$, $G : (Y^r, d_Y^r) \rightarrow (X^r, d_X^r)$ be morphisms between them. If $F \circ G \simeq \text{id}_{Y^r}$ and $G \circ F \simeq \text{id}_{X^r}$, then we say that the complexes (X^r, d_X^r) and (Y^r, d_Y^r) are **homotopy equivalent**.